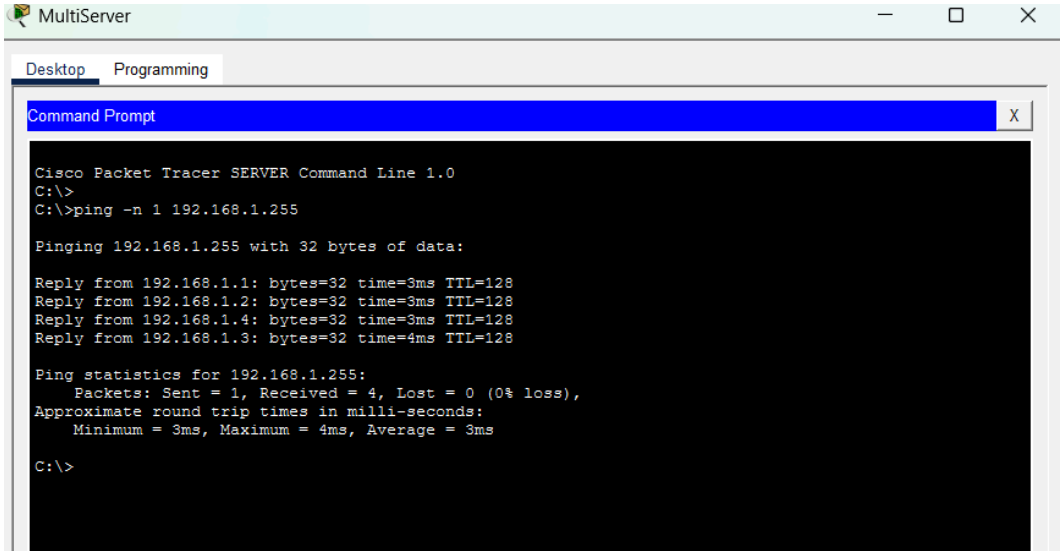
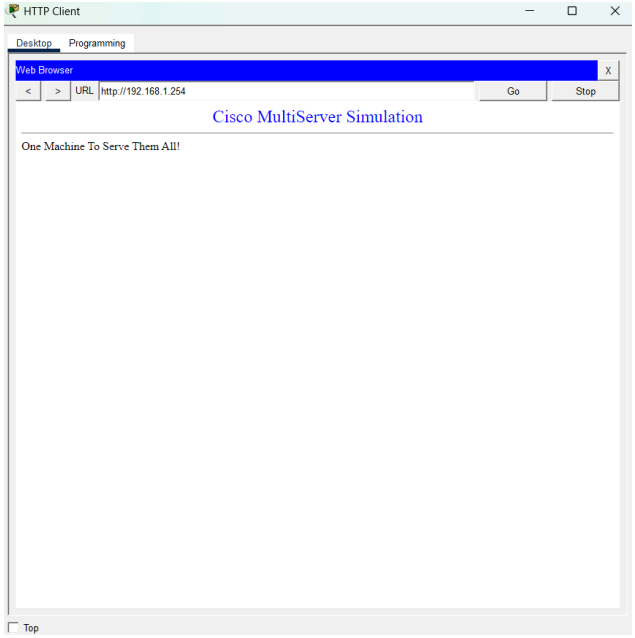
Part 1: Generate Network Traffic in Simulation Mode and View Multiplexing

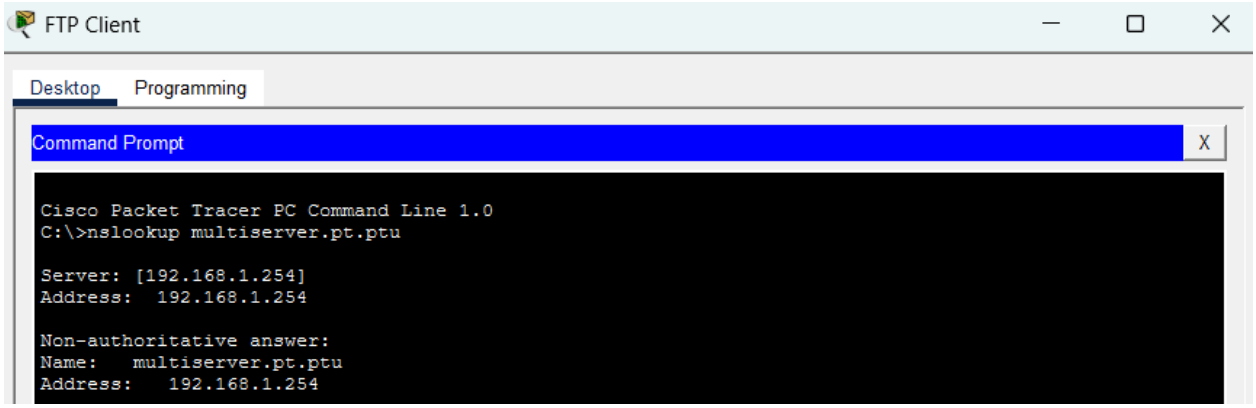
Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.



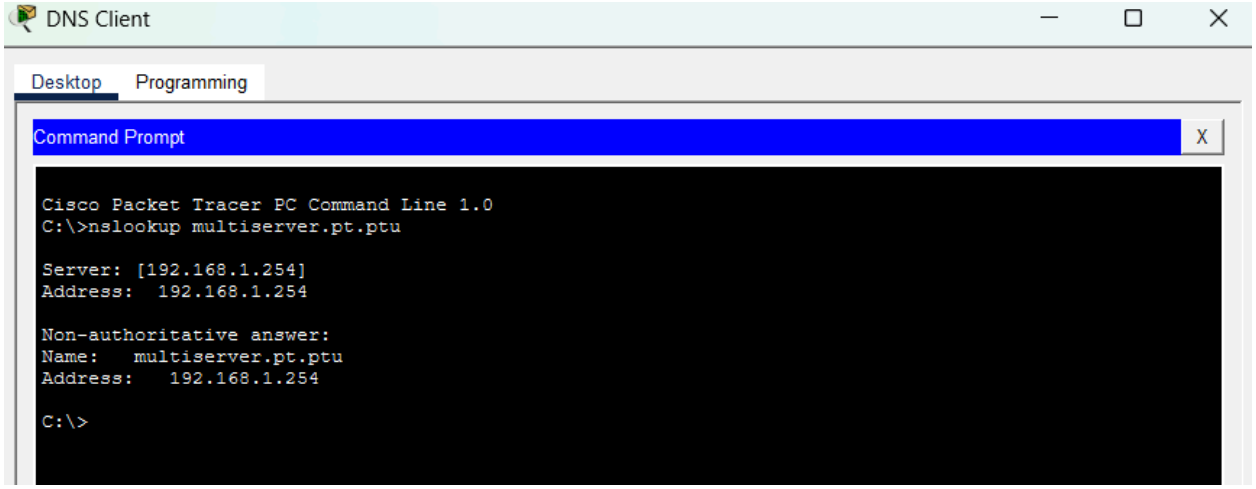
Step 2: Generate web (HTTP) traffic.



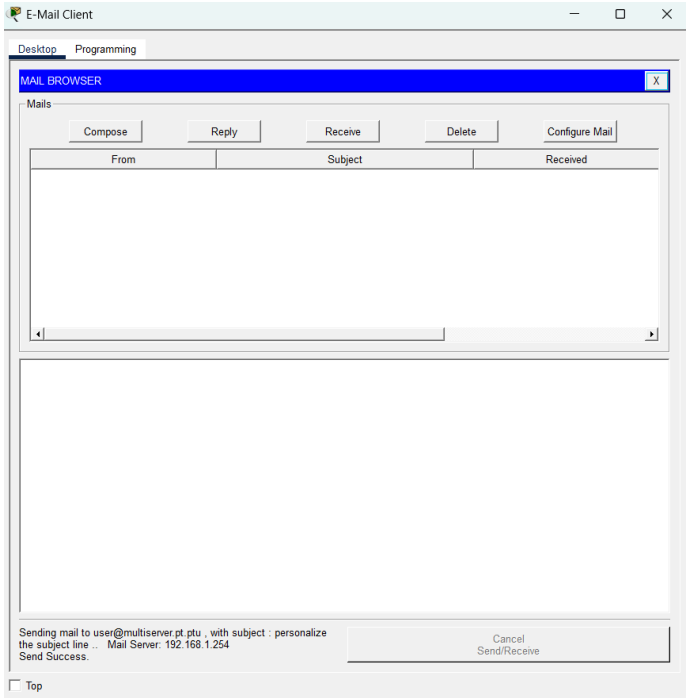
Step 3: Generate FTP traffic.

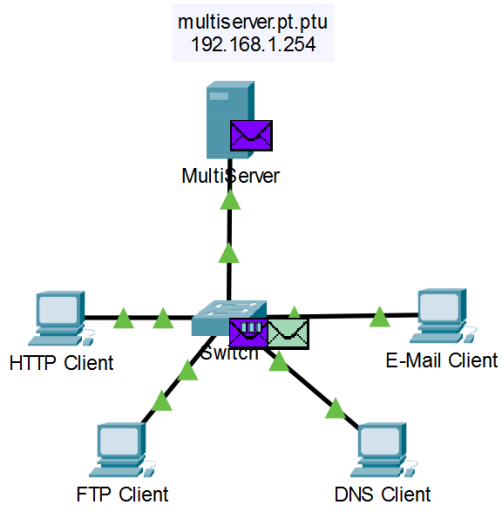


Step 4: Generate DNS traffic.



Step 5: Generate Email traffic.



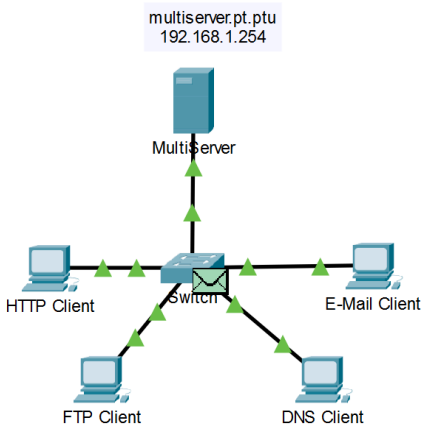


Questions:
What is this called?
Multiplexing

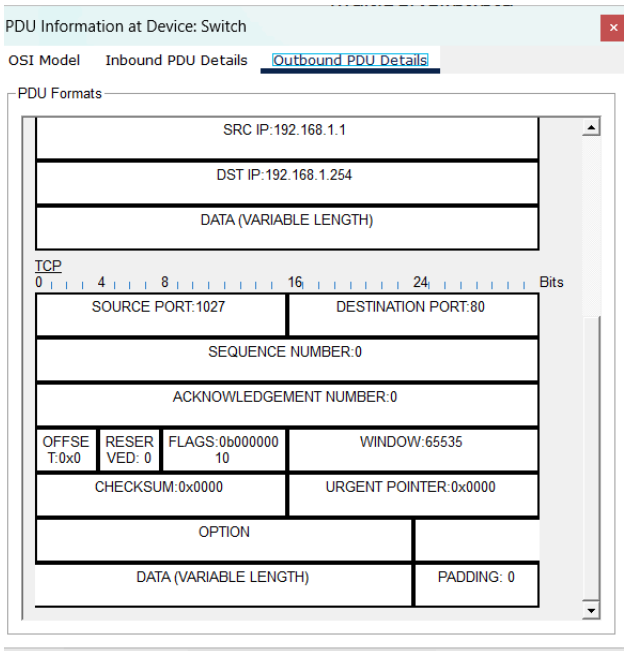
A variety of PDUs appears in the event list in the Simulation Panel. What is the meaning of the different
colors?

สีที่แตกต่างกันของ PDU ใน Simulation Panel ใช้แทน **ชนิดของโปรโตคอลที่แตกต่างกัน** เพื่อช่วยให้ผู้ใช้สามารถแยกและติดตามการทำงานแต่ละชนิดได้ง่ายขึ้นระหว่างจำลอง

Part 2: Examine Functionality of the TCP and UDP Protocols
Step 1: Examine HTTP traffic as the clients communicate with the server.



Question:
Why did it take so long for the HTTP PDU to appear?
ต้องมีการ **3-way handshake (SYN, SYN-ACK, ACK)** ก่อนเริ่มส่งข้อมูล HTTP



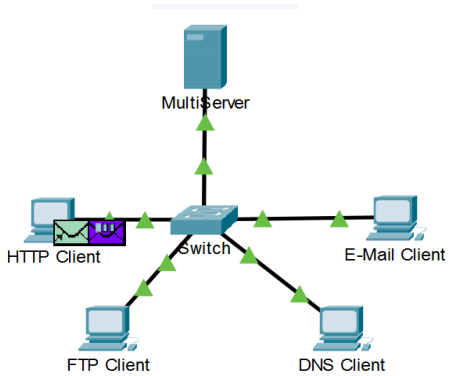
Questions:
What is the section labeled?
Transmission Control Protocol (TCP)

Are these communications considered to be reliable?

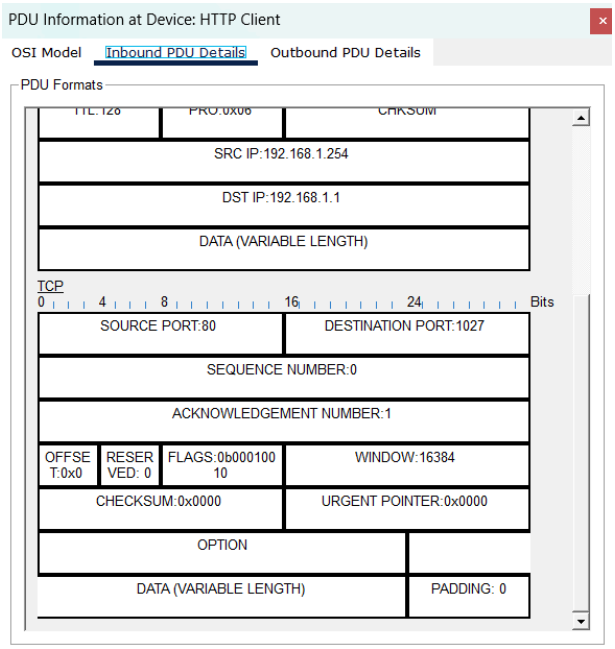
TCP เป็น reliable protocol

Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values.
SRC(1027) , DEST(80), SEQUENCE(0) , ACK NUM(0)
Flag 0b00000010
Question:
Which TCP flags are set in this PDU?
syn(1)

g. Close the PDU and click Capture/Forward until a PDU with a checkmark returns to the HTTP Client.



h. Click the PDU envelope and select Inbound PDU Details.

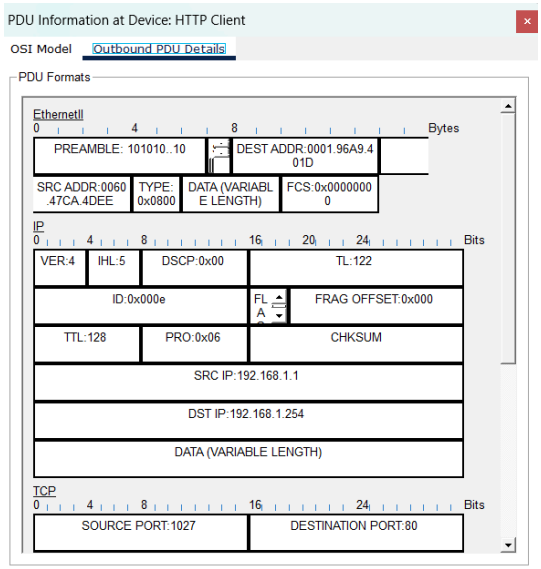


Question:

How are the port and sequence numbers different than before?

Portต้นทางและปลายทางสลับกัน, ACKเพิ่มเป็น1

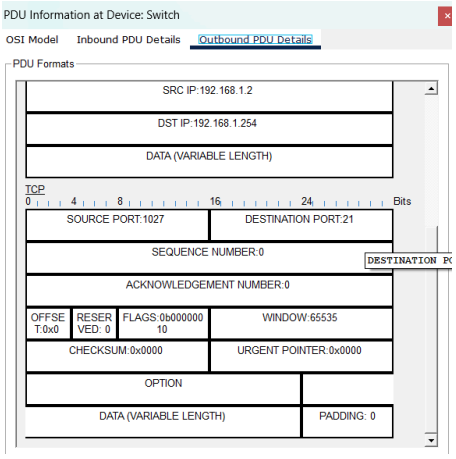
i. Click the HTTP PDU which HTTP Client has prepared to send to MultiServer. This is the beginning of
the HTTP communication. Click this second PDU envelope and select Outbound PDU Details.



Question:
What information is now listed in the TCP section? How are the port and sequence numbers different
from the previous two PDUs?
TCP เพิ่ม(SEQUENCE NUM)และ(ACK), SRC PORT สลับตำแหน่งกับ DEST PORT

j. Reset the simulation.

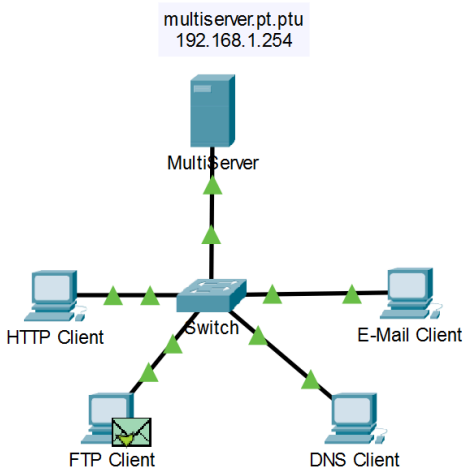
Step 2: Examine FTP traffic as the clients communicate with the server.



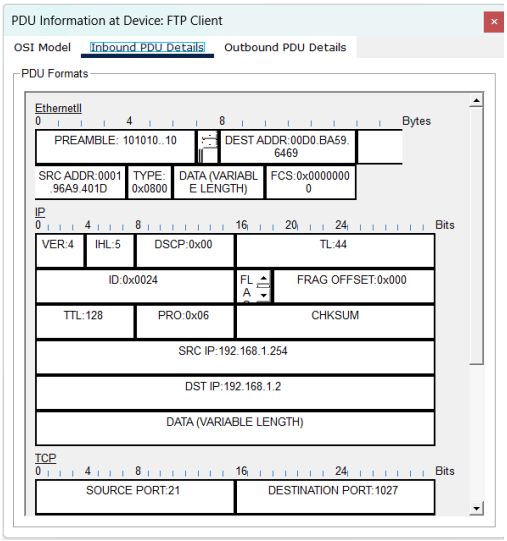
Question:
Are these communications considered to be reliable?
ใช่ มีความน่าเชื่อถือ

d. Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values.
Src:1027 , Dest: 21, seq: 0, ack: 0
Question:
What is the value in the flag field?
0b00000010

e. Close the PDU and click Capture/Forward until a PDU returns to the FTP Client with a checkmark.



f. Click the PDU envelope and select Inbound PDU Details.

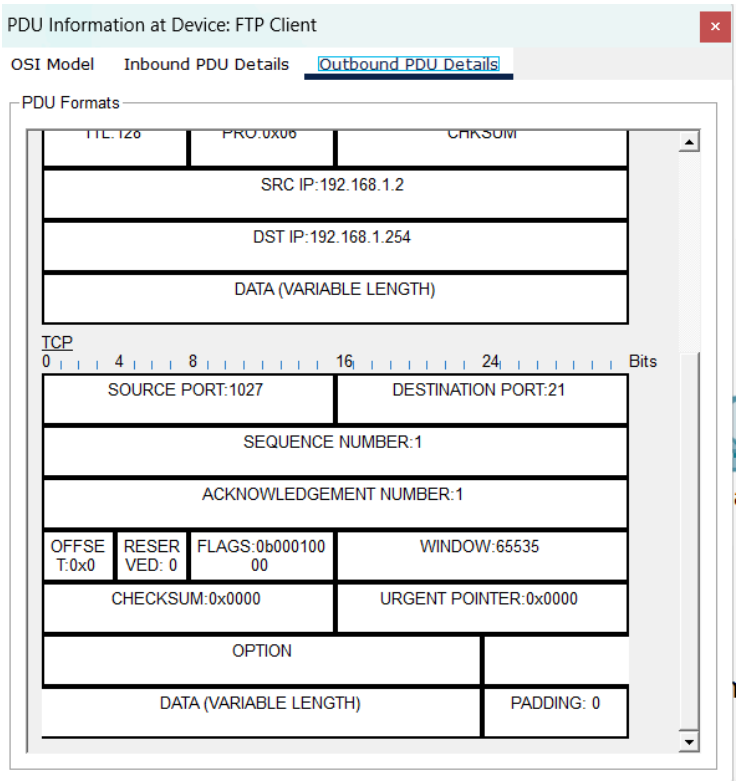


Question:

How are the port and sequence numbers different than before?

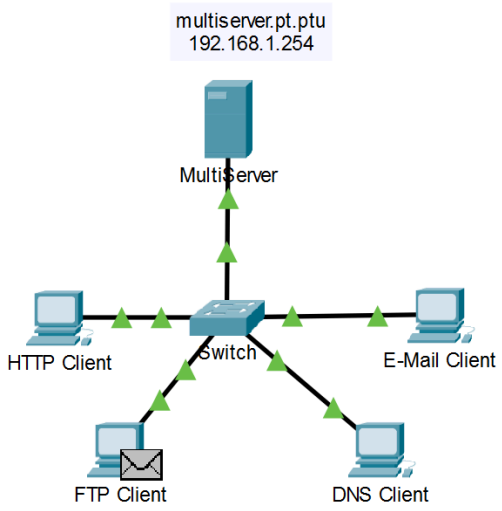
ค่าของพอร์ตสลับกัน และ(ACK)เปลี่ยนแปลงไป แต่(SEQUENCE NUM) เหมือนเดิม

g. Click the Outbound PDU Details tab.

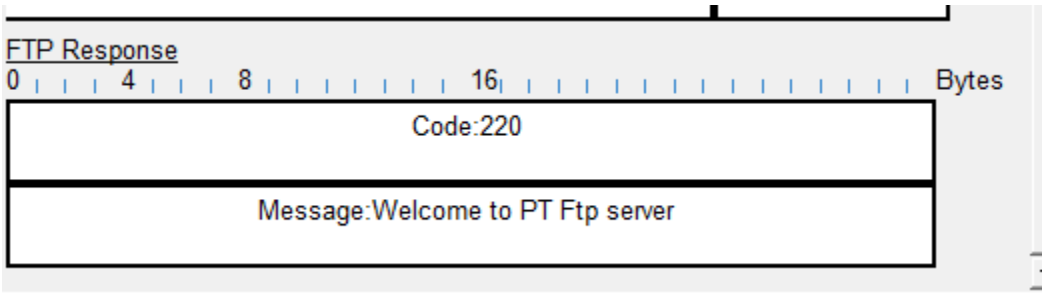


Question:
How are the port and sequence numbers different from the previous results?
ค่าของพอร์ตถูกสลับ และ(SEQUENCE NUM)และ(ACK NUM)ก็เปลี่ยนแปลงไป

h. Close the PDU and click Capture/Forward until a second PDU returns to the FTP Client. The PDU is a
different color.

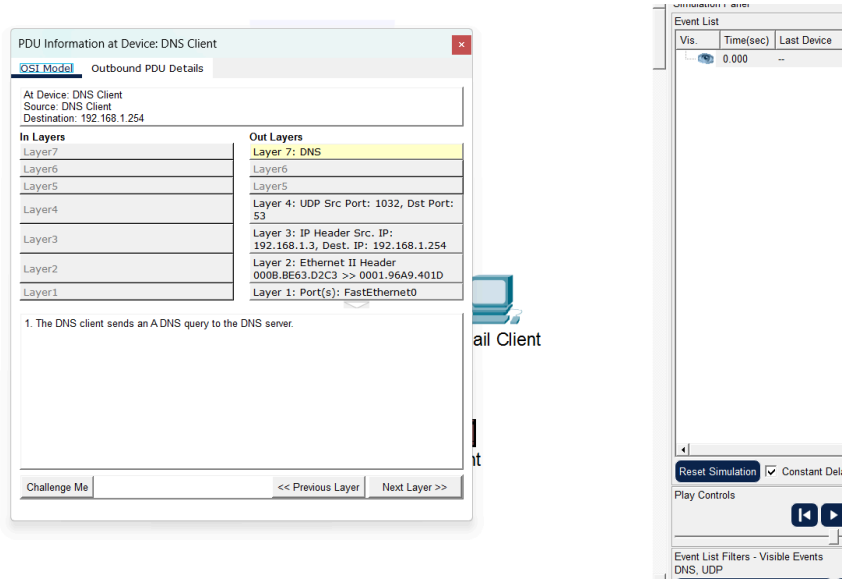


i. Open the PDU and select Inbound PDU Details. Scroll down past the TCP section.
Question:
What is the message from the server?



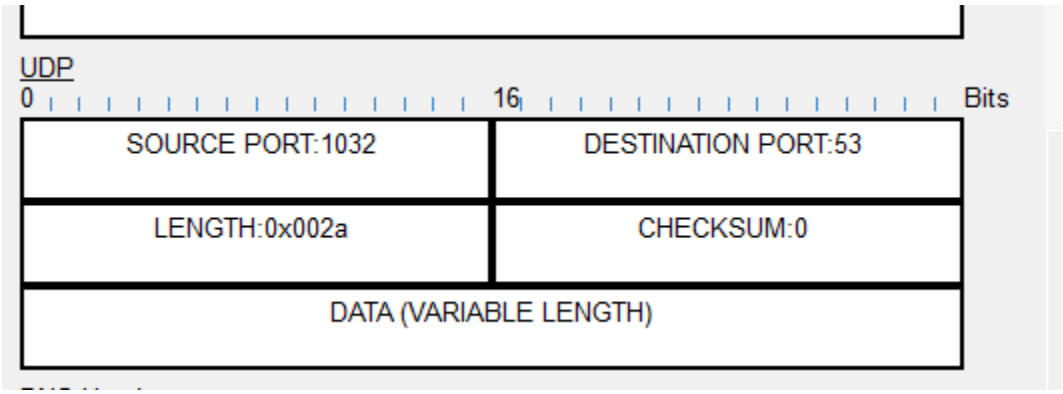
j. Click Reset Simulation.

Step 3: Examine DNS traffic as the clients communicate with the server.



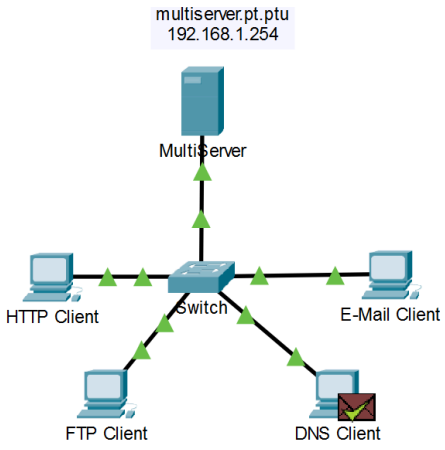
Question:
What is the Layer 4 protocol?
Src: 1025,Dst: 53

Are these communications considered to be reliable?
ใม่ ไม่หน้าเชื่อถือ
e. Open the Outbound PDU Details tab and find the UDP section of the PDU formats. Record the SRC

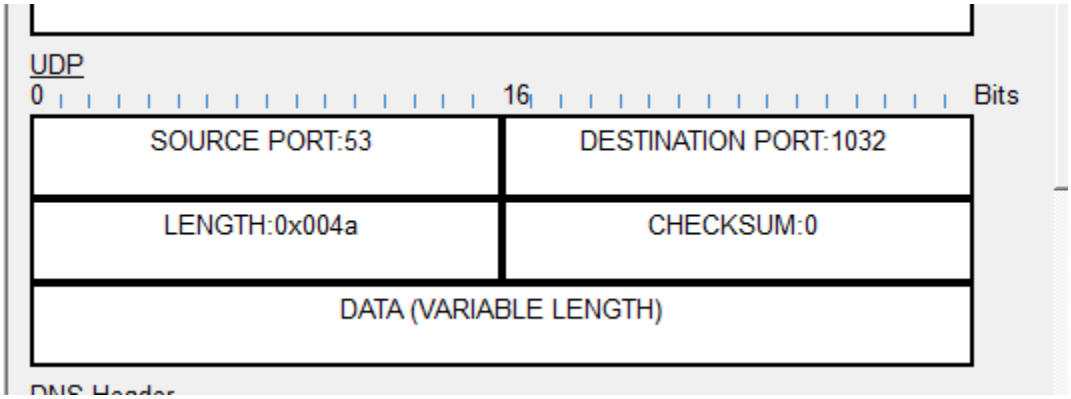


PORT and DEST PORT values.
Question:
Why are there no sequence and acknowledgement numbers?
UDP ไม่ใช้หมายเลขลำดับหรือหมายเลขยืนยัน

f. Close the PDU and click Capture/Forward until a PDU with a check mark returns to the DNS Client.



g. Click the PDU envelope and select Inbound PDU Details.

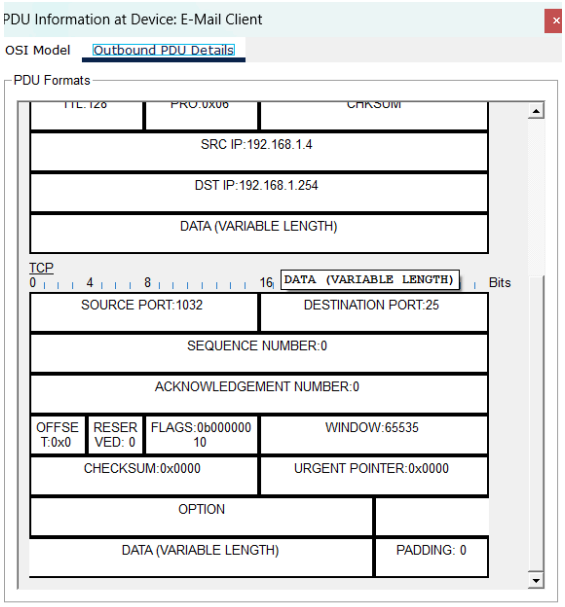


Question:
How are the port and sequence numbers different than before?
Srcสลับกับdest และlength อัพเดต
What is the last section of the PDU called? What is the IP address for the name multiserver.pt.ptu?

DNS Answer , IP:192.168.1.254

h. Click Reset Simulation.

Step 4: Examine email traffic as the clients communicate with the server.



Questions:

What transport layer protocol does email traffic use?

TCP

Are these communications considered to be reliable?

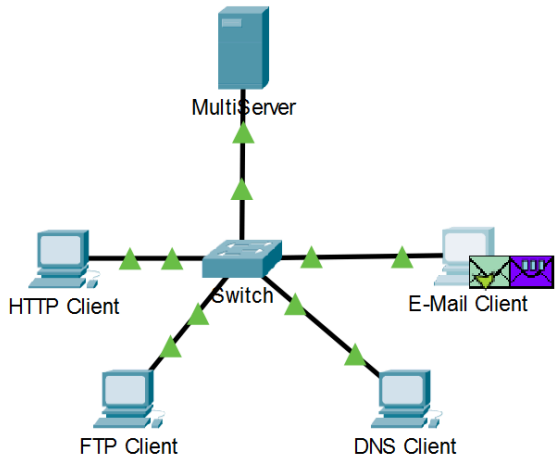
TCP เชื่อถือได้.

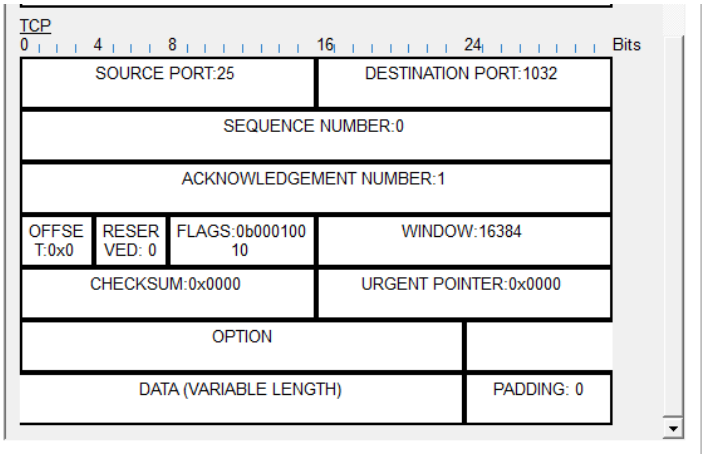
e. Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values. What is the flag field

value?

SRC PORT(1032), DEST PORT(25), SEQUENCE NUM(0), and ACK NUM(0)

f. Close the PDU and click Capture/Forward until a PDU returns to the E-Mail Client with a checkmark.





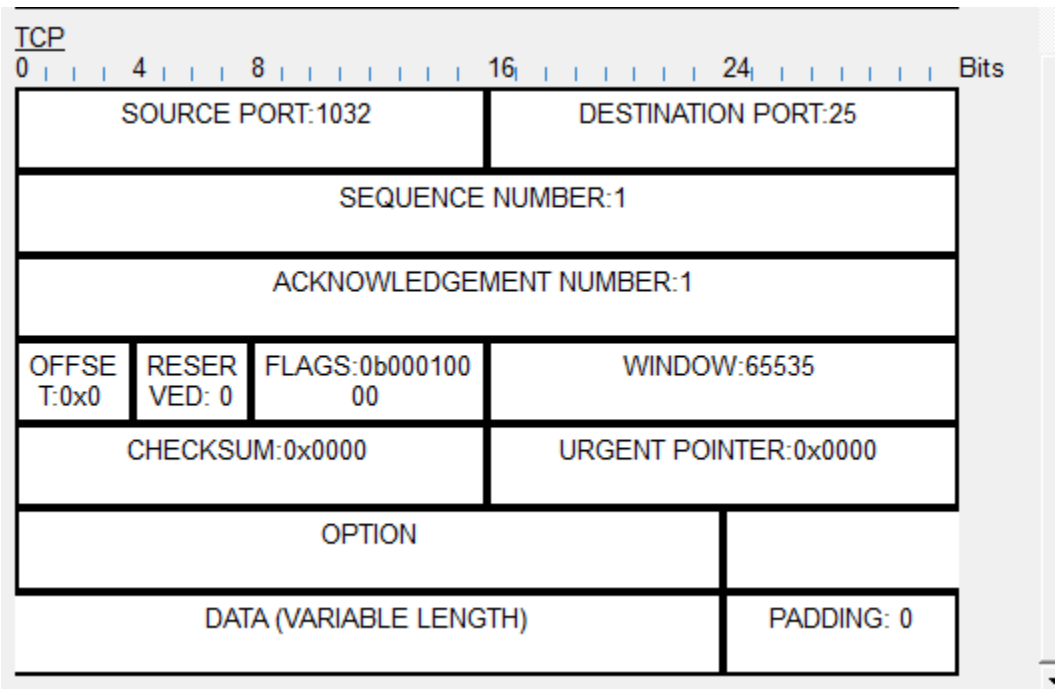
g. Click the TCP PDU envelope and select Inbound PDU Details.

Question:

How are the port and sequence numbers different than before?

ค่าของพอร์ตถูกสลับตำแหน่ง และหมายเลขยืนยันการรับส่งข้อมูล(ACK NUM) เปลี่ยนแปลงไป แต่หมายเลขลำดับ(SEQUENCE NUM)เหมือนเดิม

h. Click the Outbound PDU Details tab.

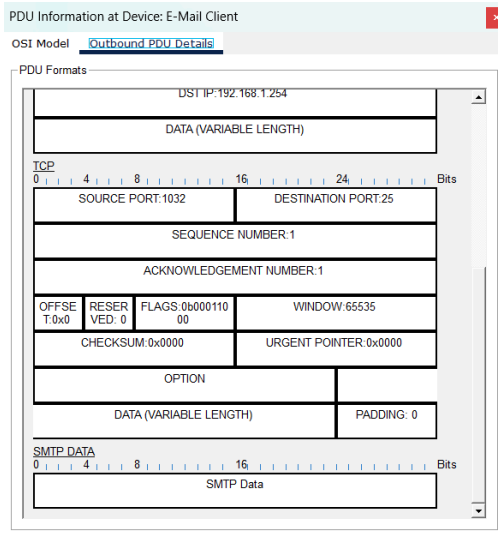


Question:

How are the port and sequence numbers different from the previous two results?

ค่าของพอร์ตถูกสลับตำแหน่ง และ(SEQUENCE NUM) เปลี่ยน

i. There is a second PDU of a different color that E-Mail Client has prepared to send to MultiServer. This
is the beginning of the email communication. Click this second PDU envelope and select Outbound PDU
Details.



Questions:
How are the port and sequence numbers different from the previous two PDUs?
เลขพอร์ต (ACK NUM) และ(SEQUENCE NUM)เหมือนก่อนหน้า

What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?

โปรโตคอลที่ใช้พอร์ต TCP 25 คือ SMTP และโปรโตคอลที่ใช้พอร์ต TCP 110 คือ POP3